

README.md

Brain Tumor Detection System

Project Overview

This project is an AI-powered Brain Tumor Detection System that leverages deep learning techniques to analyze MRI scans and detect potential brain tumors. The system provides an easy-to-use interface where users can upload MRI images, receive instant analysis, and generate a downloadable PDF report with the results.

Problem Statement

Brain tumors are life-threatening conditions that require early diagnosis for effective treatment. Traditional diagnostic methods can be time-consuming and require expert radiologists. This system aims to assist healthcare professionals by automating the detection process, increasing efficiency, and improving early diagnosis.

Features

- **Upload MRI Scans:** Users can upload MRI images for analysis.
- **AI-Powered Detection:** Uses YOLO-based deep learning models to detect brain tumors.
- **Interactive UI:** Built with Streamlit for a smooth user experience.
- **Confidence Score Analysis:** Displays detection results with confidence levels.
- **PDF Report Generation:** Generates a downloadable report for documentation and further analysis.

Technologies Used

- **Streamlit:** For building the web interface.
- **OpenCV:** For image processing and pre-processing.
- **NumPy:** For handling numerical operations.
- **YOLO (Ultralytics):** Deep learning model for tumor detection.
- **Pillow (PIL):** For handling image formats.
- **FPDF:** For generating PDF reports.

Setup and Installation

1. Create a Virtual Environment (MacOS)

```
python3 -m venv venv  
source venv/bin/activate
```

2. Install Dependencies

```
pip install -r requirements.txt
```

3. Run the Application

```
streamlit run app.py
```

Expected Outcomes

- Faster and more accurate preliminary diagnosis of brain tumors.
- Reduced dependency on manual radiology interpretation.
- Better documentation and patient management with PDF reports.

Disclaimer

This tool is designed to assist in medical diagnostics but should not replace professional medical advice. Always consult a certified medical expert for final diagnosis and treatment recommendations.